

Pore network and water retention characteristics of volcanic porous media

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Summary

Aggregate media are often characterized by multi-porous systems, which have structural and water retention characteristics that depend on the complex interaction between intra- and inter-aggregate pores. Here we investigate the structure and water retention dynamics of rigid aggregate volcanic materials. In particular, we focus on commercially used pumices, lapilli and zeolites. The aim was to estimate the air and water content through complex dual-porous systems, and thus to evaluate their suitability for vegetation growth. Both inter- and intra-aggregate characteristics were determined by means of mercury intrusion porosimetry, X-ray microtomography and water retention curves. The wilting point was determined with pressure plates, a dew point hygrometer and the sunflower method to assess their reliability at small matric potentials. Results indicate that aggregate porous media were bimodal and their heterogeneous pore network affected the water retention dynamics because (i) the large inter-aggregate pores allowed a rapid drainage near saturation and (ii) the intra-aggregate porosity held water available for root uptake and plant growth. In contrast, volcanic powders were less affected by the inter- and intra-aggregate dual-porosity. The use of a dew point hygrometer instead of pressure plates for determining small matric potentials is also suggested because pressure plates might over-estimate the water content because of poor plate and soil conductance. However, the reference potential at wilting point should be set at values greater than -1471.5 kPa (-784.8 kPa) to consider the interaction between plant roots and porous media with small hydraulic conductivity. Results from this work indicate that aggregate multi-porous media allow the simultaneous supply of oxygen and available water for plants, although the heterogeneous nature of the pore network involves uncertainties regarding water balance and root–matrix interactions.

Introduction

Concentration of the world's population in urban areas (UN, 2012) leads to a transition from a natural to anthropogenic environment and induces significant changes of land use. The covering and destruction of the soil with impervious surfaces is an intense form of 'land take' (Jones *et al.*, 2012), which causes a complete loss of water retention areas, an increase in surface water runoff and flood risks (Scalenghe & Marsan, 2009).

Artificial soils have become widespread in several fields, including horticulture, because they guarantee root aeration and drainage (Raviv *et al.*, 2002), and 'green' roof construction, because the vegetation transforms solar energy into water vapour with benefits for building insulation. With the infiltration of water into the reconstructed porous horizon, artificial soils can contribute to the reduction of impervious areas and the improvement of the urban water balance so that it approaches a more natural one (Berndtsson, 2010).

Among the substrates suitable for building artificial soils, volcanic materials are excellent for growing vegetation in a shallow profile. Pumices and lapilli are inert, porous products from explosive volcanic eruptions formed during a violent expansion of gases (Gizas & Savvas, 2007), while zeolites are volcanogenic sedimentary aluminosilicate minerals characterized by large cation exchange capacities (Ramesh & Reddy, 2011). All these materials are commercially used in agriculture, horticulture and the building industry. They are usually sorted and graded by the producer and are coarser than natural volcanic soils. Despite their incoherence and the lack of structure stability between grains, volcanic materials have gained importance as they simultaneously provide sufficient oxygen and available water for plants as well as improved texture to the soil profile.

Rigid aggregate systems often form a dual-porosity pore network that can be divided into inter-aggregate pores, surrounding aggregates and intra-aggregate pores within the aggregates (Blonquist *et al.*, 2006). Depending on the material type and size, the large inter-aggregate pores are rapidly drained and water flows through these and their contact region. The contacts are conductive

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when wet and abruptly decrease the unsaturated conductivity when drained (Carminati *et al.*, 2008). Recently, Gizas & Savvas (2007) used different pumice grades in a heated glasshouse as substrates for crop production. They reported that grain sizes significantly affected the air:water ratio and therefore the water availability for vegetation. In particular, the finer pumice (0–2 mm) was characterized by poor air-filled porosity, with the consequent restricted plant growth and yield as a result of restricted root aeration. Raviv *et al.* (1999) showed that in pumice and yellow tuff aggregates water availability was the limiting factor for plant growth, which, in turn, was mainly dependent on the unsaturated hydraulic conductivity.

Despite the widespread use of volcanic materials as substrates for vegetation growth it is not completely clear how the hydraulic behaviour, including water availability for plants, can be affected by their volcanogenic nature and grain aggregate size.

This study advances the hypothesis that the aggregate size has a major influence on the inter-aggregate porosity while the volcanogenic nature would affect the intra-aggregate porosity. Large aggregates would be dominated by the macropore fraction, resulting in poor unsaturated hydraulic conductivity and therefore a limited plant capacity to exploit the available water that is stored in the intra-aggregate porosity. On the other hand, the influence of the volcanogenic nature on hydraulic properties would be more relevant in substrates formed by powders or low-size aggregates dominated by the microporosity fraction.

A multiscale approach for the porous medium characterization was adopted for evaluating the role of macro- and micropores in water and air movement, and thus in plant growth, and quantifying the impacts of hydraulic and physical characteristics on vegetation growth through the sunflower method. We therefore analysed the structural and water retention characteristics of five different graded volcanic materials (pumices, lapilli and zeolites) by means of matric potential estimation, mercury intrusion porosimetry (MIP) and X-ray microtomography (micro-CT).

Material and methods

Materials studied

The volcanic materials used for this study were pumices, lapilli and zeolites. These graded porous media are usually employed in green roof systems for their porosity and small bulk density. They have a matrix skeleton that provides good water retention potential and exchange capabilities for the chemical elements involved in the nutrition of plants. All the materials were mined, blended and graded by a private company at different sites in central Italy. The materials were classified and tested according to their origin and grain size (Table 1).

Mercury intrusion porosimetry and X-ray microtomography

Approximately 2 g volcanic materials were air-dried and analysed by mercury intrusion porosimetry (MIP) in order to estimate the pore size diameter distribution in the range 0.0074–500 µm. Pores within the range 0.0074–15 µm were analysed with a Pascal

Table 1 Volcanic material classification and main physical characteristics

Sample code	Type of substrate	Origin	Size / mm	Bulk density / cm ³ g ⁻¹	Particle density / cm ³ g ⁻¹
P ₁ 0–3	Pumice	Site 1	0–3	0.66	2.56
P ₁ 3–8	Pumice	Site 1	3–8	0.31	2.46
P ₁ 6–14	Pumice	Site 1	6–14	0.30	2.42
P ₂ 3–7	Pumice	Site 2	3–7	0.43	2.34
P ₂ 7–12	Pumice	Site 2	7–12	0.32	2.47
L ₁ 0–3	Lapillus	Site 3	0–3	1.19	2.73
L ₁ 3–10	Lapillus	Site 3	3–10	1.18	2.77
L ₂ 0–3	Lapillus	Site 4	0–3	1.19	2.85
L ₂ 3–5	Lapillus	Site 4	3–5	1.02	2.84
L ₂ 3–10	Lapillus	Site 4	3–10	0.77	2.86
L ₂ 11–14	Lapillus	Site 4	11–14	0.84	2.84
Z ₁ 0–3	Zeolite	Site 5	0–3	0.85	2.27
Z ₁ 3–7	Zeolite	Site 5	3–7	0.82	2.24
Z ₁ 7–12	Zeolite	Site 5	7–12	0.81	2.23
Z ₂ 0–3	Zeolite	Site 6	0–3	1.10	2.45

240 porosimeter (Thermo Finnigan, Waltham, MA, USA) and those within the range 13.6–500 µm were analysed with a Pascal 140 porosimeter (Thermo Finnigan). The pore radius into which mercury (Hg) was intruded was estimated as a function of pressure using the Young-Laplace equation:

$$r = \frac{2\gamma \cos \theta}{P}, \quad (1)$$

where r is the pore radius, P is the pressure, γ is the Hg surface tension (0.47 N m⁻¹) and θ is the contact angle between Hg and aggregated materials (140°) (Hossain & Lachemi, 2006). Each measurement was replicated twice.

In order to quantify the inter-aggregate porosity and evaluate its variability between samples, volcanic materials were analysed with Skyscan 1172 X-ray microtomography (Bruker micro-CT, Kontich, Belgium). Before analysis, a cylindrical sample holder of about 100 cm³ (2.5-cm radius × 5-cm height) was filled with each type of volcanic material. According to Gabriëls & Verdonck (1991), the cylindrical sample holder was saturated and then subjected to matric potential equal to −4.91 kPa. This procedure allowed approximately the same bulk density (Table 1) to be obtained as the samples used for the estimation of water retention curves (Gabriëls & Verdonck, 1991). The X-ray micro-CT scan was set at 70 kV and 140 µA and 451 projections were collected during a 180° sample rotation at 0.4° angular incremental steps. Beam hardening artefacts were minimized with a 0.5-mm aluminium filter and averaging five identical exposures allowed noise image to be reduced. Projections were reconstructed using the cone-beam reconstruction software NRecon version 1.6.6.0 (Bruker micro-CT) to obtain a stack of at least 600 2D slices in eight-bit depths. Voxel resolution was 27 µm in all three directions. Volcanic aggregates were easily visualized and binarized with a single threshold level in order to distinguish between the inter-aggregate porosity and the volcanic materials. Total porosity, pore size distribution and specific surface

area between aggregates were calculated from each binarized stack using CTAn software version 1.12.0.0 (Bruker micro-CT). Pore-size distribution was estimated by using the sphere-fitting method (Remy & Thiel, 2002), while specific surface area (μm^{-1}) was estimated as the aggregate surface (by counting of the voxel interfaces between void and solid phases) to volume ratio.

Water retention curves

Prior to determining the water content, the disturbed volcanic samples were repacked in order to settle their bulk density. Bulk density was determined by drying a double cylinder set (each one 2.5-cm height $\times$ 6-cm diameter) filled with each substrate. According to Gabriëls & Verdonck (1991), the cylinder set was filled with the volcanic material taken to a matric potential of -4.91 kPa. After the samples were saturated, a pressure of -0.98 kPa was imposed. The rings were then separated and the lower one determined the volume (Gabriëls & Verdonck, 1991). The water retention curve (WRC) of each type of volcanic sample was calculated by means of a sandbox apparatus, pressure plates and dew point hygrometer. A total of six replicates was performed on each sample. The sandbox apparatus was used for determining water retention curves at 0, -0.25 , -0.98 , -1.96 , -4.91 and -7.85 kPa, while the water content at smaller matric potential values of -9.81 , -32.37 , -49.05 , 98.1 , -294.30 , -686.70 and -1471.50 kPa was subsequently measured by means of a pressure-plate apparatus. After the establishment of moisture equilibrium at each suction, the samples were weighed and then returned to their position to determine the water content at the smaller matric potentials. After sandbox and pressure-plate measurements, the volcanic samples were oven-dried for 24 hours at 105°C and weighed to determine the dry mass.

A hygrometer WP4-T Dewpoint Potentiometer (Decagon Devices Inc., Pullman, WA, USA) was also used for determining the matric potential in the neighbourhood of -1500 kPa. The WP4-T was first calibrated with certified 0.5 M KCl salt solution. The volcanic samples were fully saturated with distilled water for about 24 hours. They were left to evaporate into a constant temperature chamber (25°C) in plastic sample holders (1.4-cm height, 4-cm diameter). After sample equilibration, water potential was estimated with the WP4-T hygrometer. As bimodal pore distribution could be expected in this kind of material, both the classical Van Genuchten model and its dual-porosity form (Durner, 1994) were used to parameterize the WRCs with RETC, version 6.02 (Van Genuchten *et al.*, 1991).

The Van Genuchten model is as follows:

$$S_e = \frac{1}{[1 + (\alpha\psi)^n]^{1-1/n}}, \quad (2)$$

where $S_e = (\theta - \theta_r)/(\theta_s - \theta_r)$ ($\text{m}^3 \text{m}^{-3}$) is the effective saturation, θ_s and θ_r are the saturated and residual water content, respectively, θ the volumetric water content at the matric potential ψ (kPa), and α (kPa^{-1}) and n are curve fitting parameters relating to the matric head at air entry and pore size distribution, respectively.

The dual-porosity model assumed a secondary pore structure in the large range because of the presence of aggregated systems:

$$S_e = w_1 [1 + (\alpha_1\psi)^{n_1}]^{1-1/n_1} + w_2 [1 + (\alpha_2\psi)^{n_2}]^{1-1/n_2}, \quad (3)$$

where $S_e = (\theta - \theta_r)/(\theta_s - \theta_r)$ ($\text{m}^3 \text{m}^{-3}$) is the effective saturation, θ_s and θ_r are the saturated and residual water content, respectively, θ the volumetric water content at the matric potential ψ (kPa), w_i are the weighting factors for the sub-curves of the inter- and intra-aggregate pore systems ($w_1 + w_2 = 1$) and α_i and n_i are curve fitting parameters of the sub-curves relating to the matric head at air entry and pore size distribution, respectively.

The WRCs were then used to determine the pore size distribution according to the capillary model and the equivalent pore radii were calculated from the matric head by applying the Young-Laplace equation (Equation 1). The contact angle between water and aggregates was considered to be 0° and the surface tension of water at 20°C was 0.7286 N m^{-1} .

Sunflower wilting point method

As the volcanic samples were characterized by large and rigid aggregates, the root-water uptake could be affected by the root capacity to grow and interact with a heterogeneous pore network (Babalola & Lal, 1977). The sunflower method was therefore tested in all volcanic substrates in order to evaluate the plant capacity to absorb water from the volcanic medium. A total of three replicates were performed per sample. With this method, sunflowers (indicator plants) were grown in containers of uniform substrate (the volcanic materials) that were sealed to limit water loss other than that by transpiration. They were kept adequately watered until the third set of leaves appeared, at which time the watering ceased. The plants remained in an environment with a small evaporative demand until all three sets of leaves wilted. To ensure the wilting was permanent, plants were placed overnight in a humid, dark chamber. If all leaves remained wilted in the morning, the wilting point had been reached and the residual water content (θ_{wp} , g g^{-1}) was finally determined (Cassel & Nielsen, 1986). The roots were removed prior to substrate weighing to minimize the error introduced by including the roots with the substrate mass.

Statistical analysis

In order to establish whether the unimodal or the dual-porosity model (Durner, 1994) was the most suitable to describe the water retention, data were compared using a partial F test. The test established whether the increase of deviance from the dual-porosity model (D) to the unimodal (U) model was significant:

$$F = \frac{(SSQ_U - SSQ_D) / df_U - df_D}{(SSQ_D / df_D)}, \quad (4)$$

where SSQ is the sum of squares of the fitted versus the observed curves and df the degree of freedom. Water retention curves were

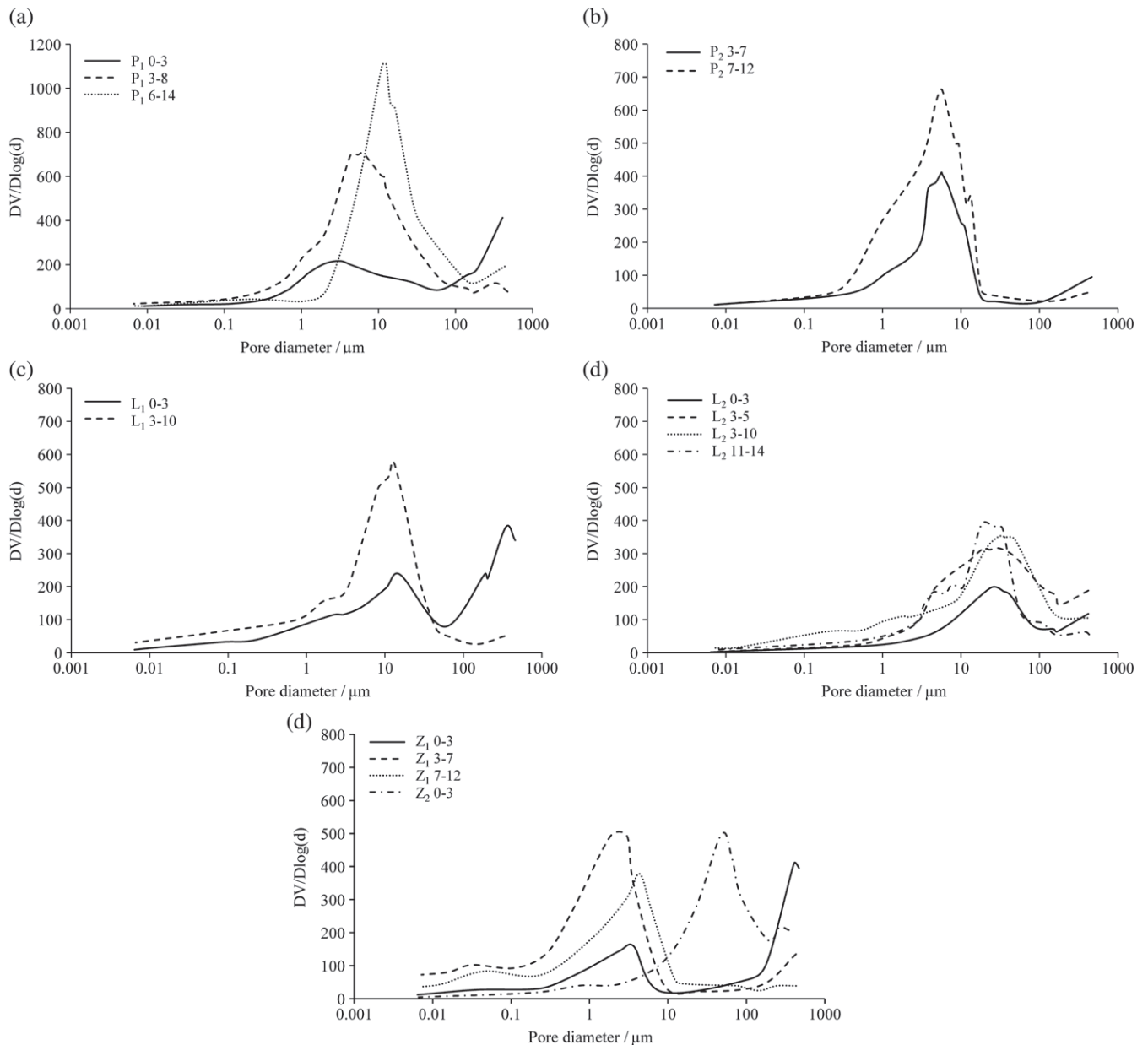


Figure 1 Pore size distribution of (a,b) pumices (P₁ and P₂), (c,d) lapilli (L₁ and L₂) and (e) zeolites (Z₁ and Z₂) of different size (mm) detected with mercury intrusion porosimetry.

also compared statistically using a partial F test. The test established whether the increase of deviance from the combined water retention curves to individual ones was significant:

$$F = \frac{(SSQ_{comb} - SSQ_{sep}) / (df_{comb} - df_{sep})}{(SSQ_{sep} / df_{sep})}, \quad (5)$$

where *SSQ* is the sum of squares of the fitted versus observed curves (*comb* = combined curves; *sep* = separate curves) and *df* the degrees of freedom (Schabenberger & Pierce, 2002). A multiple statistical significance test ($P < 0.05$) was performed with the

Bonferroni adjustment in order to compare curves of pumices, lapilli and zeolites.

Results

Mercury intrusion porosimetry and X-ray microtomography

The pore size distribution curves of volcanic aggregates, measured with mercury intrusion porosimetry (MIP), were mostly unimodal. In pumices (Figure 1) the mode was between 5.4 μm ('P₂ 7–12') and 12.5 μm ('P₁ 6–14') and between 12 μm ('L₁ 3–10') and 42 μm ('L₂ 3–10') in lapilli, while the peak was shifted markedly

towards smaller pores in zeolites (1.5 μm in 'Z₁ 3–7' and 3.5 μm 'Z₁ 7–12'). Furthermore, almost all volcanic aggregate curves had a slight increase in the pores > 300 μm . All the powders ('P₁ 0–3', 'L₁ 0–3', 'L₂ 0–3', 'Z₁ 0–3' and 'Z₂ 0–3') had a bimodal pore size distribution curve with two distinct peaks (Figure 1): the first peak was associated with pores of about 300 μm , while the second was close to 20 μm for lapilli 'L₁ 0–3' and 'L₂ 0–3', 5 μm for 'P₁ 0–3' and 'Z₁ 0–3' and 50 μm for zeolite 'Z₂ 0–3'.

Among the pumices, total aggregate porosity (Table 2) ranged between 0.56 m³ m⁻³ in 'P₂ 3–7' and 0.79 m³ m⁻³ in powder 'P₁ 3–8', while it was only 0.46 m³ m⁻³ in aggregates of lapilli with a small variability between sizes and origin (minimum of 0.42 m³ m⁻³ and maximum of 0.49 m³ m⁻³). In contrast, the powders 'L₁ 0–3' and 'L₂ 0–3' had mean values of 0.79 and 0.65 m³ m⁻³, respectively. On average, zeolites had intermediate total porosity values (0.59 m³ m⁻³).

The volcanic powders had the largest MIP ultra-macropores (500 to 100 μm), ranked as follows: 'Z₂ 0–3' (0.25 m³ m⁻³), 'P₁ 0–3' (0.21 m³ m⁻³) and 'L₁ 0–3', 'L₂ 0–3', 'Z₁ 0–3' (0.15 m³ m⁻³). Fewer ultra-macropores were found in the aggregates, ranging from a minimum of 0.028 m³ m⁻³ in 'Z₁ 7–12' to a maximum of 0.115 m³ m⁻³ in lapillus 'L₂ 3–5'.

Inter-aggregate pore size distributions, estimated by means of X-ray microtomography (micro-CT), were shifted toward greater dimensions, between a minimum of 100 μm and a maximum of 5000 μm (Table 3). The pore size class 5000 to 2000 μm was particularly the most frequent in pumice 'P₂ 7–12' (0.35 m³ m⁻³), followed by 'Z₁ 7–12' (0.31 m³ m⁻³) and 'P₁ 6–14' (0.30 m³ m⁻³). There were no pores in the same size class of powders, mainly characterized by pores in the range 500 to 100 μm (up to 100% in 'Z₂ 0–3').

Total inter-aggregate porosity (Figure 2) generally increased from small to large aggregates. For instance, inter-aggregate porosity of lapillus L₁ increased from 0.21 m³ m⁻³ in powders 0–3 mm to 0.34 m³ m⁻³ in aggregates 3–10 mm. Even inter-aggregate porosity of zeolites increased from 0.24 m³ m⁻³ in 'Z₁ 0–3' to 0.36 m³ m⁻³ in 'Z₁ 3–7' and to a maximum of 0.42 m³ m⁻³ in 'Z₁ 7–12'. Only pumice 'P₁ 0–3' had greater inter-aggregate porosity (0.53 m³ m⁻³) than larger pumice aggregates (0.40 m³ m⁻³ on average).

The MIP and micro-CT analyses overlapped in the class 500 to 100 μm . This pore class differed according to the method of analysis. It was 0.10 m³ m⁻³ with MIP and 0.15 m³ m⁻³ with micro-CT on average, but with a large variability influenced by the type and size of volcanic materials. Pumice 'P₁ 6–14' was 77% greater when calculated with MIP than with micro-CT, while the smallest difference was in powder 'Z₁ 0–3'. Large discrepancies were also observed in zeolite 'Z₂ 0–3' (84% greater in MIP than micro-CT) and pumice 'L₁ 3–10' (74% greater with micro-CT than with MIP).

Water retention curves

Almost all materials exhibited dual-porosity, therefore the dual-porosity model, with the exception of lapillus 'L₂ 0–3', described water retention better than the unimodal one ($P < 0.05$)

Table 2 Pore size distribution (m³ m⁻³) of pumices (P₁ and P₂), lapilli (L₁ and L₂) and zeolites (Z₁ and Z₂) calculated with mercury intrusion porosimetry

Sample code	MIP porosity / m ³ m ⁻³						
	Total (s.e.) ^a	500 to 100 μm (s.e.)	100 to 75 μm (s.e.)	75 to 30 μm (s.e.)	30 to 5 μm (s.e.)	5 to 0.1 μm (s.e.)	0.1 to 0.0074 μm (s.e.)
P ₁ 0–3	0.710 (0.022)	0.214 (0.006)	0.013 (0.004)	0.056 (0.011)	0.165 (0.035)	0.230 (0.018)	0.028 (0.003)
P ₁ 3–8	0.790 (0.071)	0.086 (0.028)	0.013 (0.002)	0.066 (0.014)	0.308 (0.061)	0.291 (0.020)	0.026 (0.002)
P ₁ 6–14	0.691 (0.165)	0.077 (0.029)	0.017 (0.014)	0.085 (0.052)	0.352 (0.143)	0.139 (0.072)	0.021 (0.002)
P ₂ 3–7	0.556 (0.070)	0.034 (0.010)	0.005 (0.003)	0.032 (0.022)	0.241 (0.0284)	0.228 (0.002)	0.016 (0.004)
P ₂ 7–12	0.729 (0.168)	0.019 (0.005)	0.006 (0.000)	0.032 (0.014)	0.307 (0.0387)	0.352 (0.134)	0.013 (0.005)
L ₁ 0–3	0.790 (0.040)	0.152 (0.122)	0.007 (0.006)	0.026 (0.019)	0.119 (0.0719)	0.340 (0.181)	0.145 (0.107)
L ₁ 3–10	0.420 (0.058)	0.016 (0.002)	0.006 (0.001)	0.034 (0.002)	0.204 (0.0229)	0.129 (0.003)	0.032 (0.002)
L ₂ 0–3	0.650 (0.053)	0.154 (0.030)	0.019 (0.005)	0.140 (0.007)	0.247 (0.0227)	0.082 (0.002)	0.009 (0.002)
L ₂ 3–5	0.489 (0.041)	0.115 (0.014)	0.023 (0.001)	0.095 (0.006)	0.174 (0.0203)	0.077 (0.018)	0.006 (0.000)
L ₂ 3–10	0.477 (0.069)	0.094 (0.001)	0.027 (0.003)	0.110 (0.014)	0.141 (0.0084)	0.092 (0.023)	0.014 (0.005)
L ₂ 11–14	0.443 (0.016)	0.063 (0.018)	0.023 (0.012)	0.082 (0.007)	0.152 (0.0265)	0.114 (0.041)	0.009 (0.003)
Z ₁ 0–3	0.539 (0.073)	0.150 (0.061)	0.005 (0.001)	0.012 (0.001)	0.051 (0.0023)	0.267 (0.004)	0.053 (0.003)
Z ₁ 3–7	0.536 (0.029)	0.047 (0.012)	0.002 (0.000)	0.006 (0.001)	0.058 (0.0179)	0.348 (0.002)	0.076 (0.000)
Z ₁ 7–12	0.470 (0.036)	0.028 (0.010)	0.003 (0.001)	0.015 (0.001)	0.064 (0.0244)	0.291 (0.048)	0.068 (0.003)
Z ₂ 0–3	0.819 (0.022)	0.246 (0.021)	0.092 (0.001)	0.234 (0.017)	0.135 (0.0144)	0.096 (0.023)	0.017 (0.002)

^aStandard error.

Table 3 Total porosity and pore size distribution ($\text{m}^3 \text{m}^{-3}$) of pumices (P_1 and P_2), lapilli (L_1 and L_2) and zeolites (Z_1 and Z_2) calculated with X-ray microtomography (5000 to $100 \mu\text{m}$)

Sample code	Micro-CT porosity / $\text{m}^3 \text{m}^{-3}$				
	Total porosity	5000 to $2000 \mu\text{m}$	2000 to $900 \mu\text{m}$	900 to $500 \mu\text{m}$	500 to $100 \mu\text{m}$
P_1 0–3	0.53	0.00	0.01	0.15	0.37
P_1 3–8	0.38	0.15	0.10	0.07	0.06
P_1 6–14	0.39	0.30	0.04	0.03	0.02
P_2 3–7	0.36	0.11	0.11	0.08	0.06
P_2 7–12	0.45	0.35	0.05	0.03	0.02
L_1 0–3	0.21	0.00	0.03	0.07	0.12
L_1 3–10	0.34	0.05	0.12	0.13	0.06
L_2 0–3	0.07	0.00	0.00	0.01	0.06
L_2 3–5	0.41	0.00	0.10	0.20	0.13
L_2 3–10	0.40	0.22	0.08	0.08	0.04
L_2 11–14	0.34	0.25	0.04	0.03	0.03
Z_1 0–3	0.24	0.00	0.00	0.08	0.16
Z_1 3–7	0.36	0.10	0.11	0.09	0.06
Z_1 7–12	0.42	0.31	0.04	0.03	0.04
Z_2 0–3	0.04	0.00	0.00	0.00	0.04

(Table 4). Water retention curves (WRCs) had significant differences between volcanic substrates and were affected by their origin and size ($P < 0.05$). Only the large size pumices (P_1 6–14' and P_2 7–12') and medium size lapilli (L_1 3–10' and L_2 3–5') did not demonstrate significant differences between WRCs (Figure 3).

Among the same volcanic materials, large aggregates had more water at saturation than the powders. Pumices held up to 0.87 and

$0.83 \text{ m}^3 \text{m}^{-3}$ of water in " P_1 6–14" and " P_2 7–12", respectively, lapillus ' L_2 3–10' up to $0.72 \text{ m}^3 \text{m}^{-3}$, and zeolites ' Z_1 3–7' and ' Z_2 7–12' about $0.69 \text{ m}^3 \text{m}^{-3}$. In aggregates, a slight decrease in water potential near saturation caused a sharp decrease in water content and at -32.37 kPa the water content in almost all aggregates was less than in powders.

At -1471.50 kPa water content measured by WP4-T was $0.057 \text{ m}^3 \text{m}^{-3}$ on average, with a maximum in the powder ' Z_1 0–3' ($0.095 \text{ m}^3 \text{m}^{-3}$) and a minimum in ' L_1 3–10' ($0.024 \text{ m}^3 \text{m}^{-3}$). Zeolites retained more water than pumices and lapilli on average (0.089 , 0.042 and $0.05 \text{ m}^3 \text{m}^{-3}$, respectively).

Parameters of the dual-porosity model emphasized the main differences between substrates. Generally, the weight (w) of the macropore component on WRCs increased according to aggregate size (Table 4). For example, w increased from 0.58 in the powder ' P_1 0–3' to 0.81 in the aggregates ' P_1 6–14'. The WRC parameters α_2 and n_2 , associated with inter-aggregate porosity, were greater in the aggregates than in the powders: on average α_2 was 804.42 kPa^{-1} in the former and 765.18 kPa^{-1} in the latter; n_2 was greater than 2 in large aggregate samples ' L_1 3–10', ' L_2 3–5' and P_2 7. Conversely, the distribution parameters associated with intra-aggregate porosity were, on average, 10.99 and 22.17 kPa^{-1} (α_1) and 0.92 and 1.70 (n_1) in powders and aggregates, respectively (Table 4).

Pore size distribution, estimated from WRCs according to the Young-Laplace capillary model, had two distinct peaks in almost all the samples as a result of the pore size frequency (Figure 4). A first peak was around $1000 \mu\text{m}$ in pumice, lapillus and zeolite aggregates, and a second was observed between 2 and $41 \mu\text{m}$ in ' Z_1 3–7' and ' P_2 7–12', respectively. The pumice and zeolite powders

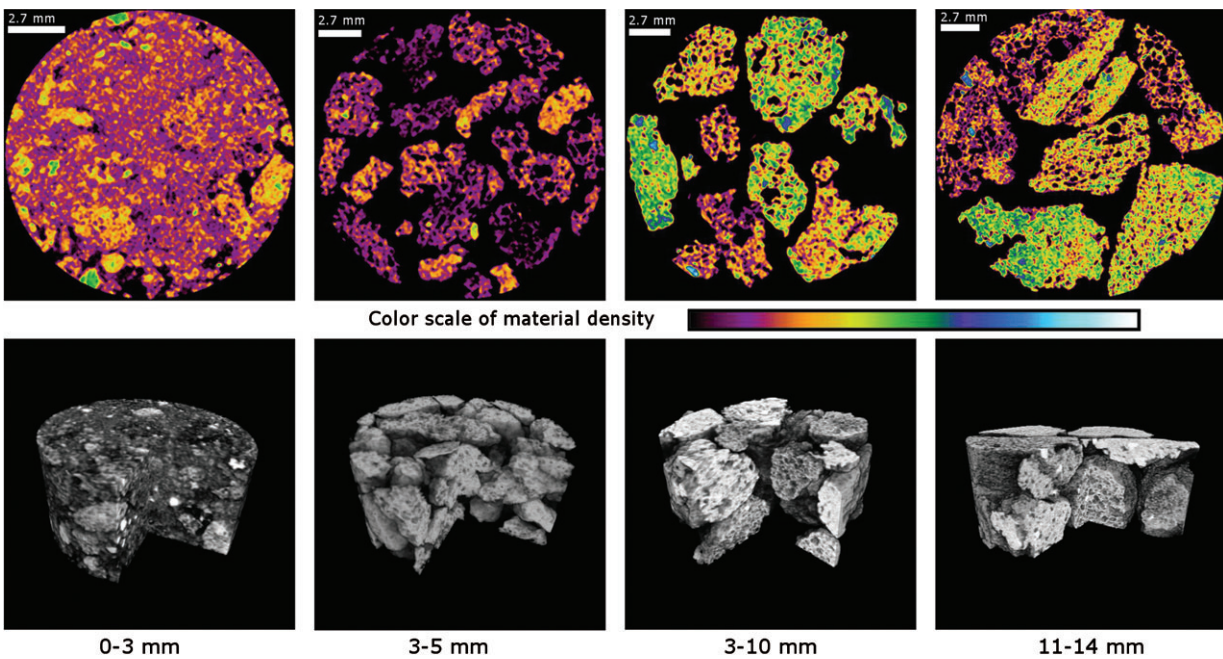


Figure 2 Inter-aggregate porosity of lapilli ' L_2 ' of different size detected with X-ray microtomography. Material density as shown on scale bar (left = empty pores; right = highly dense materials).

Table 4 Fitting parameters of the water retention curves. L₂ 0–3 was fitted according to the Van Genuchten model, while all the others were fitted according to the dual-porosity model proposed by Durner (1994)

Sample code				Intra-aggregate pores						Inter-aggregate pores				
				$\theta_r / \text{m}^3 \text{m}^{-3}$ (s.e.) ^a		$\theta_s / \text{m}^3 \text{m}^{-3}$ (s.e.)		$\alpha_l / \text{kPa}^{-1}$ (s.e.)		n_l (s.e.)		w (s.e.)		$\alpha_2 / \text{kPa}^{-1}$ (s.e.)
P ₁ 0–3	0.05	(0.00)	0.70	(0.01)	4.91	(0.98)	2.91	(0.56)	0.58	(0.06)	184.43	(24.53)	1.86	(0.21)
P ₁ 3–8	0.00	–	0.86	(0.01)	7.85	(3.92)	1.50	(0.06)	0.66	(0.09)	901.54	(289.40)	1.60	(0.26)
P ₁ 6–14	0.00	–	0.87	(0.01)	0.98	(0.98)	1.67	(0.24)	0.81	(0.05)	1137.96	(371.80)	1.51	(0.14)
P ₂ 3–7	0.02	(0.02)	0.77	(0.01)	5.89	(2.94)	1.63	(0.27)	0.64	(0.08)	738.69	(209.93)	1.84	(0.36)
P ₂ 7–12	0.00	–	0.83	(0.02)	31.39	(22.56)	1.30	(0.04)	0.55	(0.06)	426.74	(333.54)	5.22	(26.8)
L ₁ 0–3	0.00	–	0.55	(0.01)	15.70	(9.81)	1.38	(0.03)	0.76	(0.15)	146.17	(57.88)	2.40	(1.72)
L ₁ 3–10	0.04	(0.01)	0.63	(0.01)	10.79	(2.94)	1.82	(0.13)	0.52	(0.04)	482.65	(71.61)	2.59	(0.80)
L ₂ 0–3	0.00	–	0.55	(0.01)	28.45	(2.94)	1.70	(0.06)	–	–	–	–	–	–
L ₂ 3–5	0.00	–	0.70	(0.00)	19.62	(1.96)	1.47	(0.02)	0.56	(0.01)	277.62	(9.81)	4.40	(0.50)
L ₂ 3–10	0.00	–	0.72	(0.01)	5.89	(2.94)	1.37	(0.05)	0.18	(0.02)	760.28	(72.59)	1.92	(0.12)
L ₂ 11–14	0.00	–	0.70	(0.01)	7.85	(4.91)	2.88	(2.03)	1.00	(0.03)	716.13	(66.71)	1.37	(0.01)
Z ₁ 0–3	0.00	–	0.62	(0.01)	0.98	(0.00)	1.42	(0.03)	0.54	(0.02)	2413.26	(32.37)	1.77	(0.11)
Z ₁ 3–7	0.00	–	0.67	(0.01)	0.00	(0.00)	1.92	(0.47)	0.74	(0.01)	664.14	(87.31)	1.66	(0.07)
Z ₁ 7–12	0.12	(0.01)	0.70	(0.01)	0.00	(0.98)	1.14	(0.34)	0.81	(0.04)	1952.19	(782.84)	1.44	(0.10)
Z ₂ 0–3	0.00	–	0.53	(0.01)	4.91	(3.92)	1.38	(0.15)	0.78	(0.63)	316.86	(530.72)	1.36	(1.69)

^aStandard error.

had two distinct peaks at 1.2 μm in ‘Z₁ 0–3’ and at 463 μm in ‘P₁ 0–3’. Only lapillus ‘L₂ 0–3’ was unimodal and the most frequent pore size was 51 μm diameter.

Sunflower, WP4 and pressure-plate wilting point

Germination and growth of sunflower were initially delayed by the poor water supply as a result of the poor contact area between the seed and the large aggregates. However, the restricted plant growth conditions only caused a total failure of the test in pumice ‘P₁ 3–8’ (data not presented). Gravimetric water content was used in order to compare the effect of intra-aggregate pore distribution on water retention, excluding the variability related to the bulk density. Substrates were composed of different sized rigid aggregates whose disposition in the space significantly affected the bulk density and as a consequence the volumetric water content. The sunflower permanent wilting point was reached at water contents greater than those measured at –1471.50 kPa both with pressure plates and dew point hygrometer WP4-T, although regression analysis showed a better agreement for the pressure-plate technique ($R^2 = 0.72$) than WP4-T ($R^2 = 0.48$) (Figure 5a).

In general, the sunflower wilting point increased according to the aggregate size, with greater water content in coarse aggregates than in powders. For example, it was 0.17 g g^{–1} in ‘P₁ 0–3’ and 0.29 g g^{–1} in ‘P₁ 6–14’, corresponding to water potentials of –588.60 kPa in powders and –147.15 kPa in large aggregates. The same behaviour, though less pronounced, was also observed for the pressure-plate technique. Depending upon the aggregate size, increasing differences in water content at –1471.50 kPa were observed between pressure plates and WP4-T (Figure 5b). Only the powders of pumice ‘P₁ 0–3’ and lapilli ‘L₁ 0–3’ and ‘L₂ 0–3’ had smaller values with the pressure plates than with the WP4-T,

and the largest differences were observed in aggregates ‘P₂ 7–12’ (1.17), ‘P₁ 6–14’ (1.34) and ‘Z₁ 7–12’ (0.62). As a consequence, the available volumetric water content (difference between field capacity and wilting point) was affected by the method of water content estimation (Table 5). When wilting point was determined with WP4-T it was 0.13 m³ m^{–3}, followed by determination with pressure plates (0.11 m³ m^{–3}) and finally by that with the sunflower method (0.10 m³ m^{–3}).

Discussion

MIP porosity was affected mainly by the origin of the volcanic materials and most of the differences were observed between the type of medium and their origin. The artificial sieving of the materials did not significantly affect the pore size distribution. Analysis with MIP mostly explored intra-aggregate porosity because the large pores between coarse-grained aggregates were already partially filled by mercury before analysis. Only the MIP analysis on powders allowed the study of intra-aggregate porosity in the range 500 to 100 μm (Table 2). This pore class was related ($R^2 = 0.25$; $P < 0.05$) to the overlapping class measured by micro-CT techniques, although the estimation of its frequency slightly differed between the two techniques (Dal Ferro *et al.*, 2012). The MIP technique is subjected to the ‘ink bottle effect’ (Cnudde *et al.*, 2009) as it measures part of the large pores only by intruding mercury through smaller pore necks and misrepresents their size.

Both MIP and water retention curves (WRCs) identified the mode of intra-aggregate pore diameter (Figures 1, 4). The peak detected by MIP was equivalent or slightly shifted towards smaller values than that observed in WRCs. As already reported in previous studies (Stingaciu *et al.*, 2010), this difference was probably because of the influence of hysteresis, because retention data, which provide

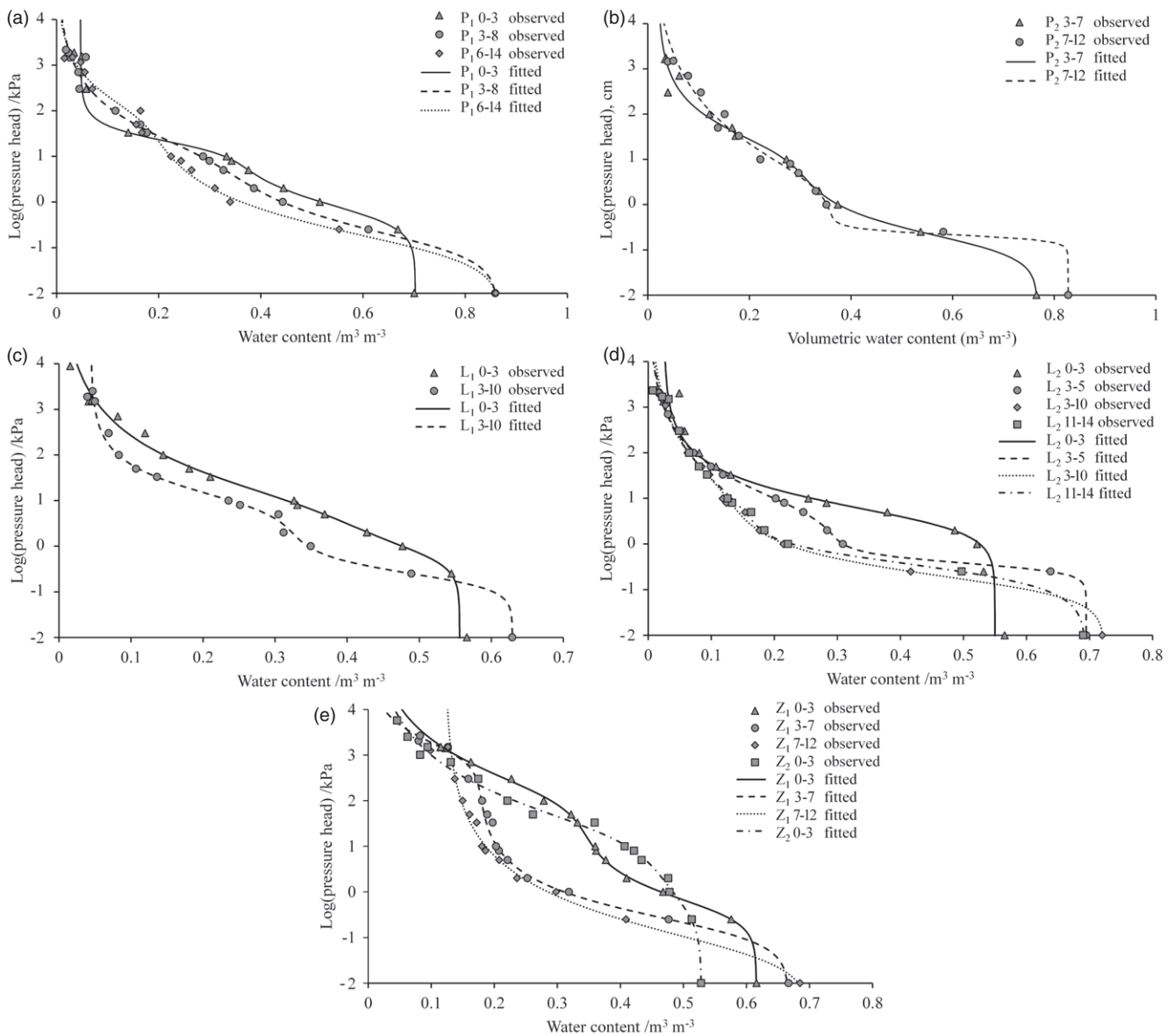


Figure 3 Water retention curves of (a,b) pumices (P_1 and P_2), (c,d) lapilli (L_1 and L_2) and (e) zeolites (Z_1 and Z_2) of different size (mm).

drainage curves, were compared with mercury data, which provide intrusion curves. Only in a few particular cases was there no relationship between MIP and WRC caused by the unimodal shape (' L_2 0–3') and lack of fit (' P_2 7–12') in WRCs at matric potential in the neighbourhood of -98.10 kPa.

Large pores between aggregates, and even the pores between particles of most of the powders (except for zeolite ' Z_2 0–3'), were visualized and quantified with X-ray micro-CT. These pores reached up to 5 mm diameter and were about 50% of total porosity on average, contributing to the rapid drainage between aggregates (Carminati *et al.*, 2008). As observed by Gizas & Savvas (2007), inter-aggregate porosity increased with the increase of aggregate size (Table 3), which in turn influenced the quantity of water and air within the volcanic media. The distinction between inter- and

intra-aggregate porosity was well conceptualized by Blonquist *et al.* (2006), who defined a critical hydraulic water content as the point where, after reducing matric potential, all inter-aggregate pores were air-filled and all the intra-aggregate pores were water saturated. The critical water content, identified by the authors as the converging point between the tangents at the inter/intra-aggregate curves' inflection points, was clearly related to the inverse of micro-pore fitting parameter α_1 . Similarly, in our volcanic substrates α_1 , which was assumed to be the inverse of air entry in the microporous system (Van Genuchten, 1980), identified the boundary between the inter- and intra-aggregate pore systems (Figure 4). Instead, for large pores between aggregates, the matric potential at air entry (α_2^{-1}) increased with the reduction in aggregate size (Table 4).

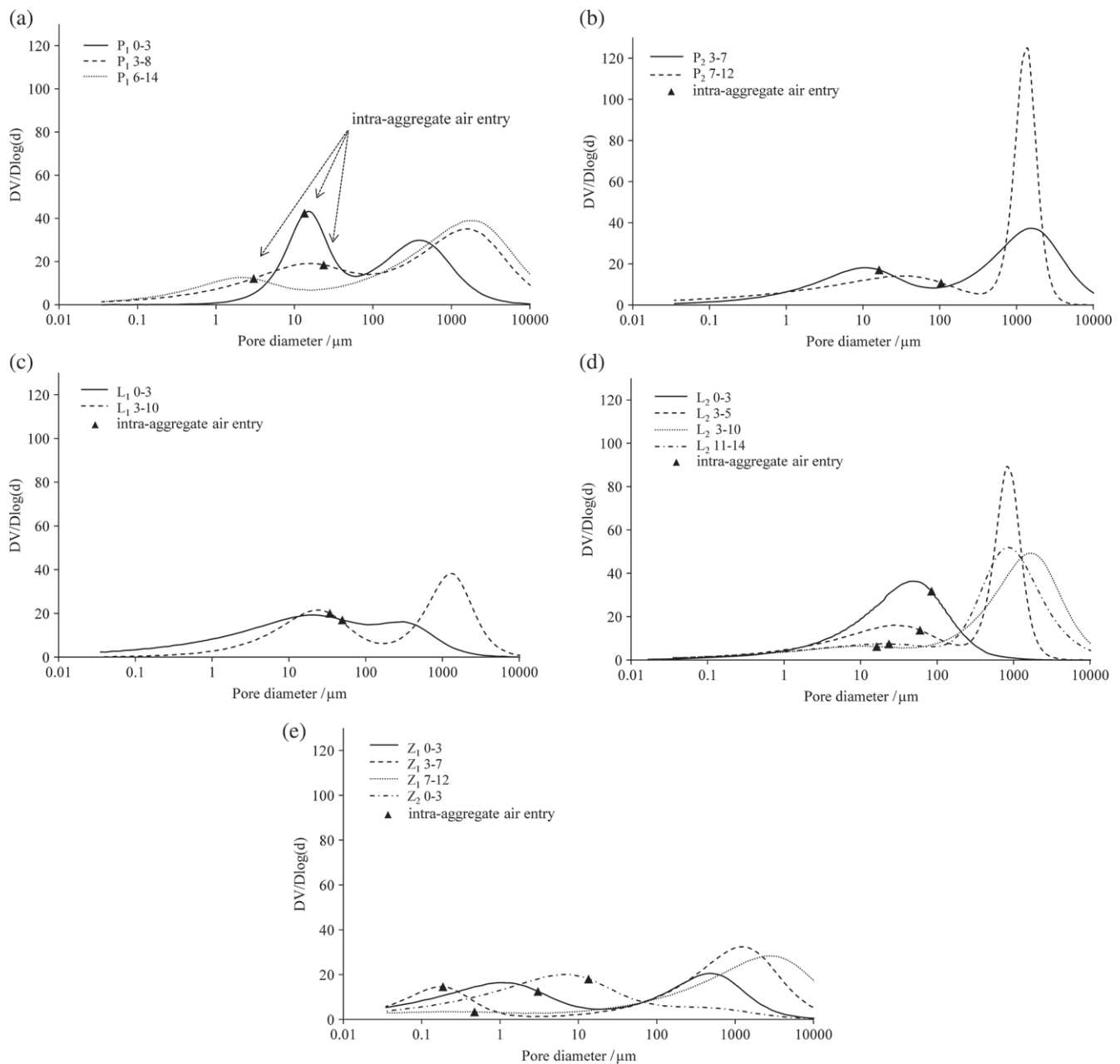


Figure 4 Pore size distribution of (a,b) pumices (P_1 and P_2), (c,d) lapilli (L_1 and L_2) and (e) zeolites (Z_1 and Z_2) of different size (mm) determined with water retention curves.

Despite the abrupt decrease of water content between aggregates, all volcanic materials retained sufficient water for plant growth because of the large intra-aggregate porosity (Table 5). Although unsaturated hydraulic conductivity was not measured, its estimation using Mualem's model showed that at -10 kPa the water flow was reduced by up to four orders of magnitude, shifting from the powders to the largest aggregates ($'L_1 0-3' = 5 \times 10^{-2} \text{ cm day}^{-1}$, $'L_1 11-14' = 7 \times 10^{-6}$, data not shown). Accordingly, Raviv *et al.* (1999) reported that in soil-less substrates (as volcanic aggregates were considered to be) water uptake by

the plants reduced the soil moisture and decreased rapidly the hydraulic conductivity at the root vicinity at a relatively large water potential.

The comparison of the available water contents within the aggregates may support this hypothesis (Table 5): the difference between AW_{SUN} and AW_{WP4} was on average $0.04 \text{ m}^3 \text{ m}^{-3}$, while that determined between AW_{PP} and AW_{SUN} was only $0.01 \text{ m}^3 \text{ m}^{-3}$, suggesting that the water limitations for plant growth were driven by the small unsaturated hydraulic conductivity. One of the strategies adopted by the plants to overcome limiting unsaturated conditions

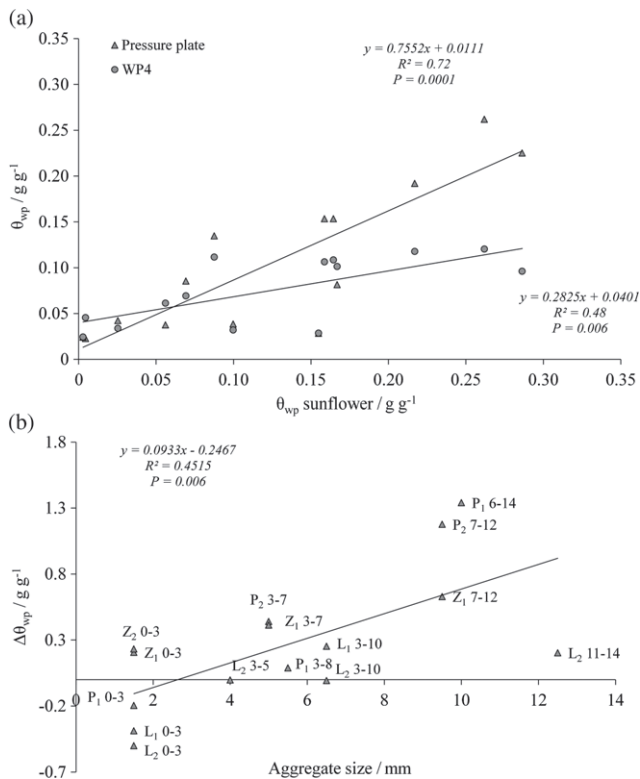


Figure 5 Relationships between (a) residual gravimetric water content (θ_{wp} , g g^{-1}) measured with the sunflower method (permanent wilting point), WP4-T (wilting point, -1471.50 kPa) ($y = 0.2825x + 0.0401$) and pressure plates (wilting point, -1471.50 kPa) ($y = 0.7552x + 0.0111$); and (b) aggregate size and $\Delta\theta_{wp}$, the difference between residual gravimetric water contents of pressure plates and WP4-T ($\Delta\theta_{wp}$ = pressure plate – WP4-T, g g^{-1}) ($y = 0.0933x - 0.2467$).

was reported by Manschadi *et al.* (2006), who studied the root system characteristics of drought-tolerant winter wheat and barley and concluded that the increase of root density and deepening are key factors under water stress conditions.

As confirmed by Carminati *et al.* (2008), who studied the unsaturated water flow through soil aggregates, the contact area of liquid films between the aggregates is crucial for water movement. The contact regions rapidly drain at larger matric potentials (about -4.91 kPa) than at the aggregate interior and become bottle-necks, leading to an abrupt decrease of unsaturated hydraulic conductivity. The authors recognized two water flow conditions: (i) close to saturation, or in the case of large infiltration rates, water can preferentially flow through the large inter-aggregate pores; and (ii) when the inter-aggregate pores are drained and most of the water is retained in the aggregates, the equivalent conductivity of a series of aggregates is equal to the aggregate conductivity scaled with the water-filled contact area. This area is inversely related to the radius of both the aggregates and their surface roughness (Carminati *et al.*, 2008).

Similarly, the size of volcanic aggregates was positively correlated with the difference in water content measured by pressure

Table 5 Available volumetric water content (AW, $\text{m}^3 \text{m}^{-3}$) estimated as the difference between field capacity (-32.37 kPa), pressure plates and WP4-T wilting point (-1471.50 kPa), and permanent wilting point by means of the sunflower method

Sample code	AW _{pp}	(s.e.) ^a	AW _{WP4}	(s.e.)	AW _{SUN}	(s.e.)
P ₁ 0–3	0.09	(0.002)	0.07	(0.002)	0.03	(0.002)
P ₁ 3–8	0.12	(0.000)	0.15	(0.000)	–	–
P ₁ 6–14	0.10	(0.004)	0.14	(0.004)	0.08	(0.004)
P ₂ 3–7	0.11	(0.005)	0.13	(0.005)	0.10	(0.005)
P ₂ 7–12	0.10	(0.017)	0.14	(0.017)	0.10	(0.017)
L ₁ 0–3	0.17	(0.005)	0.14	(0.005)	0.14	(0.005)
L ₁ 3–10	0.09	(0.006)	0.10	(0.005)	0.11	(0.006)
L ₂ 0–3	0.10	(0.003)	0.08	(0.003)	0.13	(0.003)
L ₂ 3–5	0.09	(0.001)	0.09	(0.001)	0.12	(0.001)
L ₂ 3–10	0.08	(0.001)	0.08	(0.004)	0.00	(0.001)
L ₂ 11–14	0.06	(0.006)	0.07	(0.006)	0.01	(0.000)
Z ₁ 0–3	0.22	(0.005)	0.24	(0.005)	0.26	(0.005)
Z ₁ 3–7	0.07	(0.035)	0.11	(0.002)	0.06	(0.026)
Z ₁ 7–12	0.02	(0.028)	0.08	(0.002)	0.00	(0.021)
Z ₂ 0–3	0.26	(0.002)	0.28	(0.002)	0.28	(0.002)

^aStandard error.

AW = available volumetric water content; subscripts PP, WP4 and SUN refer to wilting points determined with pressure plates, WP4-T and the sunflower method, respectively.

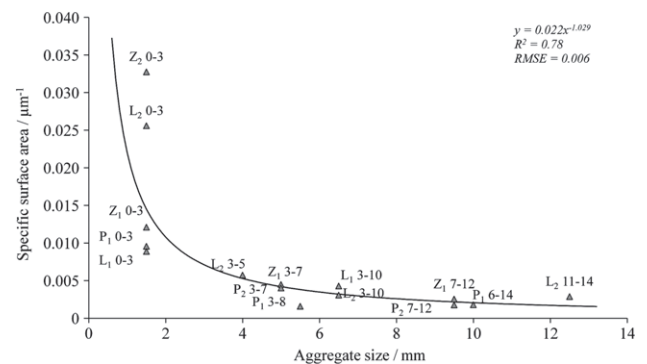


Figure 6 Relationship between aggregate size (mm) and specific surface area (μm^{-1}), calculated by means of X-ray microtomography.

plates and WP4-T (Figure 5b) at -1471.50 kPa. This behaviour suggested that in pressure plates the unsaturated hydraulic conductivity of aggregates was extremely small, requiring a very long time for equilibration and practically leading to lack of equilibrium (Gee *et al.*, 2002; Solone *et al.*, 2012). The surface area of aggregates decreased according to their size (Figure 6) and this might lead to a contraction of the water-filled contact area. No such reduction was observed with WP4-T measurements because the method is based on the equilibration of the vapour phase.

The substrate capacity to sustain plant growth was demonstrated by the sunflower method, which had a permanent wilting point that showed comparable results with those obtained with pressure plates

(Figure 5a). Likewise, Tolk (2003) indicated a correspondence between water content at -1471.50 kPa and wilting point estimated by the sunflower method. Results clearly suggested that hydraulic conductivity was a driving factor in the wilting process (Raviv *et al.*, 1999). It is probable that hydraulic conductivity decreased in large aggregates and water content at wilting point increased accordingly.

Conclusions

A thorough knowledge of the physical and hydraulic properties of volcanic materials is needed in order to evaluate their potential use for growing vegetation and reducing the impact of impervious surfaces. It was shown that the porosity of rigid volcanic aggregates was bimodal and a dual-porosity approach was suitable for describing the water movement. The intra-aggregate porosity was mainly affected by the origin of volcanic materials, while the aggregate size did not change their pore characteristics. The aggregate structure of zeolites was substantially different from pumices and lapilli, with pores of an order of magnitude smaller than the other substrates. However, the main differences were observed at inter-aggregate porosity because drainage within the pores and the water content at wilting point increased according to the increase of aggregate size, indicating an abrupt reduction of unsaturated conductivity in large aggregate materials. These findings suggest that the use of methods based on the vapour phase equilibration is suitable for determining WRC at small matric potentials. However, the reference potential at wilting point should be set at values greater than -1471.50 kPa (-784.80 kPa) to consider the interaction between plant roots and porous media with small hydraulic conductivity. Nevertheless, failures observed in the sunflower experiment demonstrated that plants might be exposed to additional severe limitations in the early stages of their growth (for example during germination) caused by the small surface area of aggregates. The capacity of roots to absorb water by direct contact with the large aggregates represents an additional element of uncertainty regarding water dynamics in volcanic media and must be investigated in the future.

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